Enhancing Balinese Gamelan Classification using a CNN with Multi Feature Fusion

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***Abstract*— As an intangible cultural heritage of Indonesia, Balinese gamelan possesses a richness that has evolved thru the Wayah, Madya, and Anyar eras. For preservation, musicological analysis, and education, the automatic classification of this historical period of gamelan is very important. This research proposes a system for classifying the historical periods of Balinese gamelan using a Convolutional Neural Network (CNN) supported by multi-domain audio feature fusion. MFCC is used to capture the essence of timbre, Mel Spectrogram is used as a time-frequency representation, and Discrete Wavelet Transform (DWT) is used to perform multi-resolution analysis. A series of test scenarios were tested to compare the performance of single-feature CNN models, traditional ML models (such as SVM, Random Forest, and Bayesian), and different feature fusion strategies. The test results show that the feature fusion approach consistently outperforms single-feature CNN and traditional ML models. The best CNN model used early fusion of three features (MFCC, Mel Spectrogram, and DWT), achieving an accuracy of 91.15%. This result demonstrates that fusing complementary audio features with a CNN architecture is a highly effective and robust solution for the complex task of traditional music classification.**

Keywords—classification, gamelan, MFCC, mel-spectogram, DWT

I. Introduction

Balinese gamelan music is an important part of Indonesian culture. It is full of life and energy. There are three main historical periods in its evolution: Wayah (old/ancient tradition), Madya (middle/transitional tradition), and Anyar (new/modern tradition) [1]. The periodization into Wayah, Madya, and Anyar fundamentally reflects the evolution of the gamelan barungan (ensemble) in terms of instrumentation and function. For instance, the transition from bilahan-dominated ensemble in Wayah to the inclusion of kendang and pencon (knobbed gongs) in Madya dramatically changes the acoustic profile, making the music more dynamic and rhythmically complex, and introducing phenomena like ngumbang-ngisep (acoustic beating)." In musicology, one of the most important tasks is to introduce and classify gamelan musical works into the right historical periods. This helps with preservation, comparative analysis, and a better understanding of this cultural heritage[1].

Musicology experts usually do this kind of classification by listening carefully and understanding the context in which the music is played. This process is very useful, but it takes a lot of work, is subjective, and depends on having experts available. The swift progress in Music Information Retrieval (MIR) and Deep Learning (DL) technologies has created opportunities for the development of a Balinese gamelan classification system. According to [2], DL is now the most common method for classifying audio. Convolutional Neural Networks (CNNs) have shown to be very good at recognizing different audio patterns because they can learn hierarchical feature representations directly from the data, especially from time-frequency representations like spectrograms [3],[4].

However, classifying gamelan is hard because of its unique features. Audio recordings of gamelan, especially historical ones, often have different levels of quality and noise. So, audio enhancement methods like spectral subtraction are useful for making the signal better before classification, as shown in the study of recognizing Gamelan instruments [5]. Specifically for Balinese gamelan, the challenge in capturing the sound's unique acoustic qualities, such as the sharp transient attacks of the metal instruments and the ngumbang-ngisep beating phenomenon. While MFCC and Mel Spectrogram capture general timbre and spectral energy, they may not sufficiently delineate the rapid percussive dynamics. Discrete Wavelet Transform (DWT), known for its ability to provide better time resolution for high-frequency transients, is therefore essential to complement the spectral features and robustly capture the percussive elements. This study integrates three components: Mel-Frequency Cepstral Coefficients (MFCC) for timbral attributes [6], Mel Spectrogram for time-frequency information [3], and Discrete Wavelet Transform (DWT) for transient and multi-resolution analysis, pertinent for elucidating the percussive elements of gamelan. A hybrid methodology integrating DWT and spectrogram has demonstrated efficacy in music genre classification [7].

This study suggests employing a complementary integration of three audio feature typesvMel-Frequency Cepstral Coefficients (MFCC), Mel Spectrogram, and Discrete Wavelet Transform (DWT) to encapsulate the richness of acoustic information in gamelan music. MFCC is known for being a reliable way to show timbre characteristics. The Mel Spectrogram shows how the spectral energy changes over time on a perceptual Mel scale. This is often turned into an image-based representation so that DL models can use it. The DWT, on the other hand, lets you look at things at different resolutions and is sensitive to transients and temporal textures at different scales. It is believed that combining the three types of features will create a more detailed and accurate audio representation for CNNs. This will make it easier to tell the difference between Balinese gamelan historical periods (Wayah, Madya, Anyar) than if you used the features on their own. The feature fusion method has worked well in a number of areas of audio classification, including environmental sound classification and medical sound classification. The

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primary contributions of this research encompass: (1) the design and implementation of an optimized CNN model for the classification of Balinese gamelan periods. (2) A thorough comparison of how well different features and fusion strategies work, which shows that three-feature fusion (MFCC, Mel Spectrogram, DWT) is the best.

II. Related Work

Automatic music classification, including traditional music like gamelan, has become a growing research focus as MIR and deep learning have advanced.

A. Classification of Gamelan Music and Other Traditional Music

Computational studies on gamelan music have covered several areas. In study [8], Short-Time Fourier Transform (STFT) and Deep Learning (DL) were used for the transcription of Demung gamelan music. Another study [5] applied spectral subtraction for audio enhancement before gamelan instrument classification using MFCC and K-Nearest Neighbors (KNN), demonstrating the importance of preprocessing for gamelan data. Another study successfully classified the tempo of the kendang, a gamelan instrument, using MFCC-CNN with high accuracy [6]. However, the classification of Balinese gamelan based on its historical period using multi-domain feature fusion remains an area that requires further exploration.

B. Audio Feature Representation

Feature selection is a crucial step in audio classification [2]. Various studies have compared the effectiveness of different features. Spectrogram vs. Waveform Representation: Most DL models for audio use time-frequency representations like Mel Spectrogram as input for CNNs [2], [4]. This approach transforms 1D audio signals into 2D "images," enabling the use of proven strong CNN architectures. However, there is also growing interest in end-to-end models that learn directly from raw waveforms to avoid potential information loss during the feature extraction process [9], [10]. A comparative study on false singing detection showed that spectrogram-based models generally still outperform waveform-based models [11]. MFCC is a standard feature for timbre [6]. The Mel Spectrogram is popular as a "image" input for CNNs [4]. DWT, thru wavelet analysis, effectively captures temporal and transient information, making it complementary to the spectrogram. A hybrid approach combining DWT and spectrogram has proven successful in music genre classification [7]. The fusion of these features forms the basis of this research methodology.

C. Feature Fusion

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III. Material and Method

The research methodology is systematically designed to analyze and classify Balinese gamelan recordings into the Wayah, Madya, or Anyar historical periods using CNN with the fusion of MFCC, Mel Spectrogram, and DWT features.

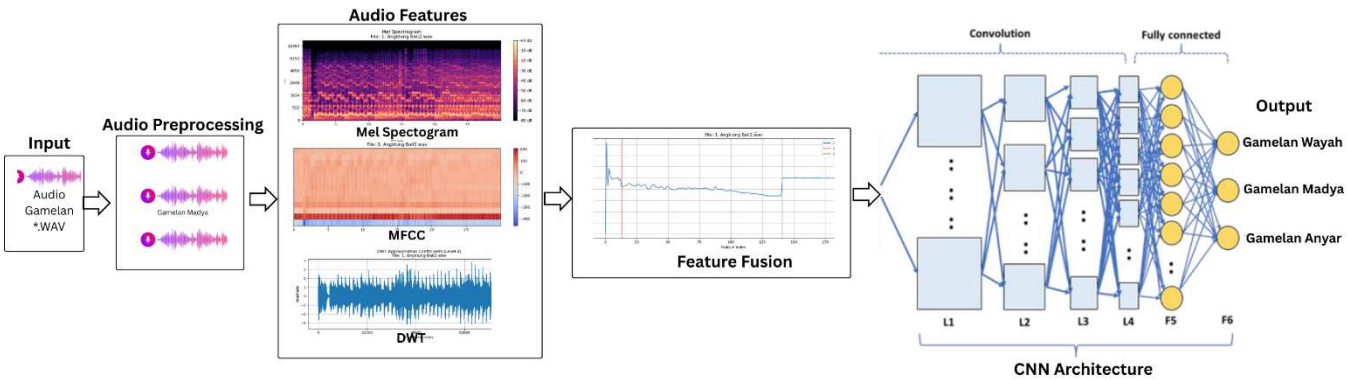

Fig. 1. Research Methodology

A. Dataset and Preprocessing

The dataset comprises 595 audio recordings of Balinese gamelan music obtained from web sites like www.youtube.com, www.spotify.com, and www.joox.com. Each recording is categorized into three classes based on its historical period (Wayah, Madya, Anyar) as per the book [1]. There are 196 audio files for the gamelan wayah class, 199 audio files for the gamelan madya class, and 167 audio files for the gamelan wayah class. All audio files are converted to mono WAV format with a sample rate of 44.1 kHz. Amplitude normalization is executed to standardize the volume levels across recordings [14]. Extended recordings are segmented into brief 30-second chunks.

B. Audio Features

Three features were retrieved from each audio segment: Thirteen Mel-Frequency Cepstral Coefficients (MFCC) have been extracted. The outcome is a 2D matrix (number of coefficients x number of frames) that is subsequently averaged (np.mean) along the temporal axis to get a singular 13-

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dimensional feature vector [15]. Mel Spectrogram: A dB-scaled Mel Spectrogram was generated using 128 Mel bands. Analogous to MFCC, the resultant matrix was averaged along the temporal axis to get a 128-dimensional feature vector. DWT analysis was conducted with the 'db4' wavelet to a depth of four levels. The approximation coefficients from the final level are extracted and standardized to a length of 64, resulting in a 64-dimensional feature vector.

C. Feature Fusion

This study employs an initial feature fusion approach to develop a holistic audio representation for the CNN model. This methodology integrates three synergistic components: Mel Spectrograms (128-dimensional vectors) that encapsulate perceptual spectral texture, MFCCs (13-dimensional vectors) that denote the distinctive timbral fingerprint of the sound source, and DWTs (64-dimensional vectors) that delineate the fundamental structure of the signal. The three distinct feature vectors are subsequently concatenated into a singular 205-dimensional hybrid vector. This unified representation offers a comprehensive audio description to the model, concurrently incorporating details like texture, timbre, and structure. The objective is to enhance the discriminative capacity of the features and, ultimately, augment the accuracy and generalization proficiency of the model in classification tasks.

D. Test Scenario

This research utilizes an early feature fusion strategy to create a comprehensive audio representation for the CNN model. This approach combines three complementary features: Mel Spectrograms (128-dimensional vectors) that capture perceptual spectral texture, MFCCs (13-dimensional vectors) that represent the unique timbre fingerprint of the sound source, and DWTs (64-dimensional vectors) that extract the fundamental structure of the signal. These three different feature vectors are then concatenated into a single 205-dimensional hybrid vector. This combined representation provides a holistic audio description to the model, simultaneously integrating information about texture, timbre, and structure. The goal is to improve the discriminative power of the features and, ultimately, increase the accuracy and generalization ability of the model in classification tasks.

E. Convolutional Neural Network (CNN) Architecture

The proposed CNN architecture consists of several main blocks, drawing inspiration from successful architectures in audio classification. The CNN architecture used comprises several convolutional blocks (including Conv2D layers, MaxPooling2D, Dropout, and ReLU activation), followed by a Flatten layer and a Dense layer for final classification with softmax activation.

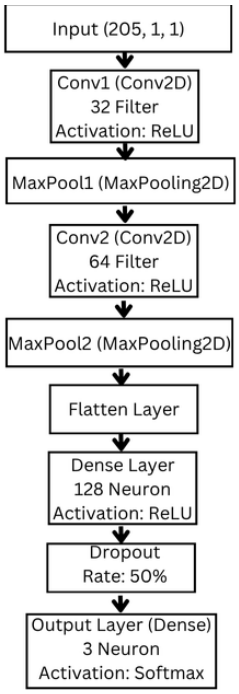

Fig. 2. CNN Architecture

The Convolutional Neural Network (CNN) architecture proposed in this study is designed for classification and is divided into two main stages: feature extraction and classification, as illustrated in Figure 2. The feature extraction stage consists of two sequential convolutional blocks. The first block uses a Conv2D layer with 32 filters and ReLU activation function, followed by a MaxPooling2D layer to reduce spatial dimensions. The second block continues this process with a deeper Conv2D layer (64 filters, ReLU activation) and a second MaxPooling2D layer to capture more complex features. The extracted features are then flattened by a Flatten layer before entering the classification stage. This stage begins with a Dense (fully connected) layer with 128 neurons (ReLU activation) for high-level pattern learning. To prevent overfitting, a Dropout layer with a rate of 0.5 was applied. In conclusion, a Dense layer with 3 neurons and the Softmax activation function is used as the output layer to generate classification probabilities for each target class [16].

F. Model Training and Validation

The CNN model training and validation process begins with the feature-extracted dataset being divided into training data (80%) and training validation, test data (20%) using the train_test_split function from scikit-learn. The stratify parameter is used to ensure equal class proportions in the training and test sets. The model was compiled using the Adam optimizer, the categorical_crossentropy loss function suitable for multi-class classification, and the accuracy metric. The model was trained with a batch size of 32. The training is set to run for a maximum of 100 epochs. To prevent overfitting and find the optimal stopping point, the EarlyStopping mechanism is implemented. Training will stop if the validation loss does not show improvement for 10 consecutive epochs (patience=10). The model with the best weights from the training process will be restored (restore_best_weights=True) for final evaluation.

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IV. Result and Discussion

The experimental results from various test scenarios, based on the data, are summarized in table I and Table 2.

TABLE I. Results of Traditional Machine Learning Classification

Classification Method	Feature	Accuracy (%)
Traditional Baseline Machine Learning		
SVM	MFCC	84.07
Random Forest	MFCC	81
Bayesian	MFCC	73
SVM	Mel Spectrogram	83
Random Forest	Mel Spectrogram	77
Bayesian	Mel Spectrogram	53

TABLE II. CNN Classification Result

Classification Method	Feature	Accuracy (%)
Single Feature Baseline (CNN)		
CNN	MFCC	74.79
CNN	DWT	73.95
CNN	Mel Spectrogram	81.51
Two Feature Fusion (CNN)		
CNN	MFCC-DWT Initial Fusion	81.42
CNN	MFCC-DWT Final Fusion	75
CNN	MFCC-DWT Fusi Hybrid	74.34
Three Feature Fusion (CNN) -		
CNN	MFCC-Mel Spectrogram-DWT Initial Fusion	91.15
CNN	MFCC-Mel Spectrogram-DWT Final Fusion	86.73
CNN	MFCC-Mel Spectrogram-DWT Hybrid Fusion	82.3

A. Results Analysis

According on the test findings presented in Tables 1 and 2, the suggested Feature Fusion: The paramount finding from these data is the unequivocal dominance of the three-feature fusion model. The CNN model, utilizing a mixture of MFCC, Mel Spectrogram, and DWT, achieved an accuracy of 91.15%, surpassing all other methodologies. This substantiates the premise that integrating information from diverse domains timbre (MFCC), perceptual spectrum shape (Mel Spectrogram), and transient dynamics (DWT) yields the most effective representation for CNN models. The Function of Fusion in Enhancing Performance: The combination of two features (MFCC-DWT) achieving an accuracy of 81.42% indicates promise for enhancement, nearly matching the performance of Mel Spectrogram alone at 81.51%. A notable enhancement in performance is observed with the inclusion of Mel Spectrogram in the fusion, affirming its status as a robust anchor feature, whose efficacy is amplified when integrated with additional complementing features. A hybrid methodology integrating wavelet-based features (DWT) and spectrograms has demonstrated efficacy in music genre categorization tasks [7].

The performance of traditional machine learning models revealed that the Support Vector Machine (SVM) utilizing MFCC features had an accuracy of 84.07%. This model not only exceeded the performance of existing machine learning models but also surpassed convolutional neural networks utilizing single MFCC (74.79%) and spectrogram (73.95%) features. This suggests that for feature sets with reduced dimensions, such as MFCC, robust classical models like SVM can be exceptionally competitive. Nonetheless, its limits are evident when confronted with high-dimensional features or when juxtaposed with sophisticated deep learning fusion methods, where SVM significantly underperforms relative to the 91.15% accuracy. Comparison of a Singular Feature in CNN: Among the individual features, the Mel Spectrogram (81.51%) demonstrated the highest efficacy for CNN, succeeded by MFCC (74.79%) and DWT (73.95%). This aligns with extensive research in the domain of MIR [4], which has demonstrated that time-frequency representations on the Mel scale are particularly effective for processing by CNN architectures, owing to their resemblance to human auditory perception and image-like configuration.

With an accuracy rate of 91.15%, this strongly indicates that a deep learning technique employing a well-structured feature fusion strategy is preferable for complex gamelan music classification compared to methods utilizing single features or traditional machine learning models.

B. Accuracy and Loss Analysis

Figures 3 and 4 illustrate the model's performance curves during the training process, depicting accuracy and loss metrics in relation to the number of epochs for both training and validation datasets. The visual examination of these two curves yields profound insights into the dynamics of model learning and its capacity to generalize to novel input.

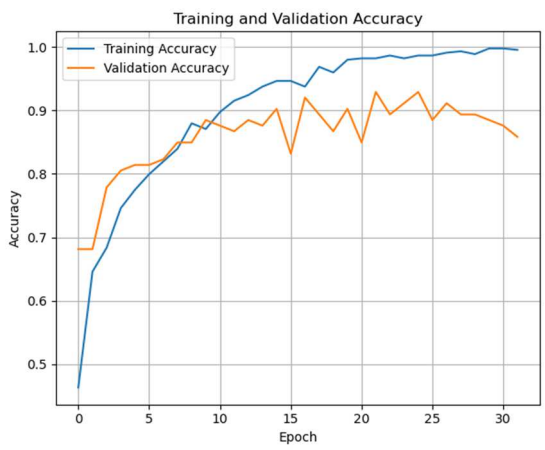

Fig. 3. Training and Validation Accuracy

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Fig. 4. Training and Validation Loss

Figures 3 and 4 indicate that the model effectively assimilated the training data, demonstrated by the training accuracy nearing 100% and the loss approaching zero. The optimal performance on new validation data was attained at approximately epoch 15, with maximum accuracy surpassing 90%. Subsequent to this ideal juncture, a split emerged between the training and validation curves: validation accuracy appeared to plateau while validation loss commenced an upward trajectory. This occurrence unequivocally signifies overfitting, as the model starts to "memorize" the training data, resulting in diminished generalization capability. This image demonstrates the model's superior performance and underscores the significance of the Early Stopping method to preserve the optimal model (about at epoch 15) prior to performance deterioration from overfitting.

C. *Confusion Matrix Analysis*

Figure 5 illustrates the Confusion Matrix of the optimal model, which is the Convolutional Neural Network (CNN) employing early fusion of MFCC, Mel Spectrogram, and DWT features.

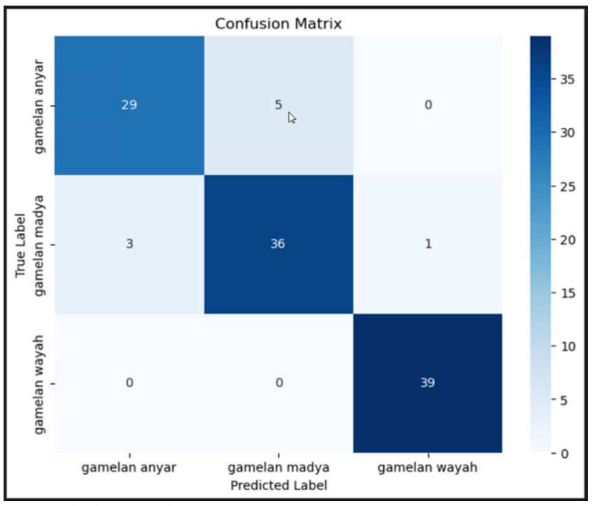

Fig. 5. Confusion Matrix

The examination of the Confusion Matrix reveals that for the "Gamelan Anyar" class, the model accurately recognized 29 cases, whereas 5 examples were incorrectly labeled as "Gamelan Madya". The "Gamelan Madya" class exhibited robust performance, with 36 examples accurately classified, despite a few misclassifications: 3 instances were incorrectly identified as "Gamelan Anyar" and 1 instance as "Gamelan Wayah". The highest performance was observed in the "Gamelan Wayah" class, where all 39 occurrences were accurately identified without any misidentification to other classes. The pattern in this Confusion Matrix experimentally corroborates the model's accuracy of 91.15% documented in Table 1. The occurrence of False Negatives between "Gamelan Anyar" and "Gamelan Madya" suggests ambiguity or resemblance in their acoustic properties, necessitating further investigation for future model optimization.

V. Conclusion

This research effectively revealed that employing a Convolutional Neural Network (CNN) utilizing feature fusion of MFCC, Mel Spectrogram, and Discrete Wavelet Transform (DWT) is an efficacious method for identifying Balinese gamelan according to its historical period (Wayah, Madya, Anyar). The suggested three-feature fusion model attained an accuracy of 91.15%, markedly surpassing both single-feature models and conventional machine learning models. This outcome highlights the necessity of amalgamating supplementary information from several audio feature domains to tackle the intricacies of conventional music. Future work should explore Late Fusion strategies combined with Attention mechanisms. The current early fusion aggregates feature vectors globally; an Attention layer, however, could be implemented on the CNN's feature maps to allow the model to automatically weigh the importance of specific temporal segments

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